

MAT 1341A - Test 5, 2014

17 November, 2014.

Instructor: Barry Jessup. **Phone contact – extension: 3536.**

Family Name:_____

First Name:_____

Student number:_____

Multiple choice answers →

For the marker's use only →

1	
2	
3	
subtotal	
4	
5	
6	
[Bonus] 7	
Total	

PLEASE READ THESE INSTRUCTIONS CAREFULLY.

1. Read each question carefully, and **answer all questions in the space provided after each question.** For questions 4 to 7, you may use the backs of pages if necessary, but be sure to indicate to the marker that you have done this.
2. Questions 1 to 3 are multiple choice. These questions are worth 1 point each and no part marks will be given. Please record your answers in the space provided above.
3. Questions 4 and 5 are worth 6 points each, question 6 is worth 4 points, and part marks can be earned in each. **The correct answers here require justification written legibly and logically: you must convince the marker that you know why your solution is correct.** Question 7 is a bonus question and is worth only 2 points. It is much more difficult to earn points in the bonus question.
4. Where it is possible to check your work, do so.
5. Good luck! Bonne chance!

1. If $C = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, and D is a $3 \times n$ matrix then the second row of the matrix CD is

A. the same as the second row of C.

B. the same as the first row of D.

C. the same as the second row of D.

D. the sum of the first and the second rows of D.

E. the sum of the first and the third rows of D.

F. the sum of the second and third rows of D.

2. Find all values of x for which $(1, 0, 1), (-1, 2, 1)$ and $(1, x, 3)$ are linearly independent.

A. For $x = 2$ only

B. For all x not equal to 2

C. $x = 0$ or $x = 1$

D. For all x

E. For no x

F. For $x = 0$ only

3. Suppose $A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Which one of the following statements is true ?

A. A^{-1} does not exist.
B. The third row of A^{-1} is $[0 \quad 0 \quad 1]$.
C. The second row of A^{-1} is $[-1 \quad 1 \quad 0]$.
D. The first row of A^{-1} is $[1 \quad 1 \quad -1]$.
E. The second column of A^{-1} is $[0 \quad 1 \quad -1]^t$.
F. All of B, C, D, E are true.

4. Let

$$A = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ -2 & 2 & 1 & 2 \end{bmatrix}.$$

(a) Find a basis for the row space of A.

(b) Find a basis for the column space of A.

(c) Find a basis for $\ker A = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

(d) Find the dimension of $\{Ax \mid x \in \mathbf{R}^4\}$.

5. Let $u_1 = (1, 1, 0, 0), u_2 = (-1, 1, 0, 0), u_3 = (0, 0, 1, -1)$, and $U = \text{span}\{u_1, u_2, u_3\}$.

a) Show that $\{u_1, u_2, u_3\}$ is an orthogonal set.

b) Briefly explain why $\{u_1, u_2, u_3\}$ is a basis of U.

c) Find the best approximation to $(1, -1, 2, -1)$ by vectors in U.

d) Extend $\{u_1, u_2, u_3\}$ to a basis of $\mathbf{R}^4$.

6(a). State whether the following is true or false, and justify your answer.

If A and B are 4×4 matrices such that the rows of AB are linearly independent, then the columns of A are linearly independent.

6(b). Let A be a $n \times n$ matrix with real entries. Give three additional statements equivalent to

"The rank of A is n

(i)

(ii)

(iii)

7. [Bonus] Suppose W is a subspace of $\mathbf{R}^4$ with $\dim W = 3$. Prove carefully that there is at least one vector $u \in \mathbf{R}^4$, with $||u|| = 1$, and which is orthogonal to every vector in W. How many such vectors are there?

(You may use this page for rough work or solutions that did not fit on previous pages.)